

# HEMANYA SAINI

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## PROFILE

Electronics and VLSI Engineering student at VIT Vellore with hands-on experience in FPGA design and embedded systems. Known for bridging hardware and software - equally comfortable working with digital systems and pitching to non-technical stakeholders. Seeking opportunities in technology, product, or operations roles where analytical thinking and strong communication can contribute to real-world projects.

## EDUCATION

**B.Tech in Electronics and Communication Engineering (VLSI Design & Technology)** 2024 - 2028

Vellore Institute of Technology (VIT), Vellore, Tamil Nadu · CGPA: 3.65 / 5.0

**CBSE Class XII - Science Stream** 2023

Ajanta Public School, Gurugram, Haryana · Score: 71.4%

**CBSE Class X** 2021

Ajanta Public School, Gurugram, Haryana · Score: 93.8%

## SKILLS

**Coursework** Basic Electrical & Electronics Engineering · Computer Architecture · Digital Systems Design · Electromagnetic Field Theory · Electronic Circuits · Electronic Materials · Microprocessors & Microcontrollers · Physics of Semiconductor Devices · Scripting Languages & Verification · Semiconductor Device Modelling

**Languages** HDL (Verilog, SystemVerilog) · TCL · Perl · Python · C · Java · MATLAB

**EDA Tools** Sentaurus TCAD · Intel Quartus Prime · Siemens Questa · Keil uVision · Multisim

**Hardware** FPGA (Altera DE2-115, AMD PYNQ-Z2) · ARM Cortex-M / STM32 · Raspberry Pi · ESP32

**Soft Skills** Public Speaking & Debate · Technical Writing · Research & Analysis · Marketing & Outreach · Cross-functional Teamwork

## EXPERIENCE

**Vice President of Substantive Affairs & Substantive Trainer - Legal and Military** Jan 2026 - Present

VITMUNSOC, Vellore Institute of Technology

- Presiding over delegate and substantive training for a 2,000-member society across 6+ committee genres (international law, defence policy, strategic affairs), raising research and debate standards across all committees.

**Under Secretary General (Sponsorships) - VITMUN '26** Oct 2025 - Present

Vellore Institute of Technology

- Secured sponsorships and managed outreach for VITMUN 2026, coordinating communication across all committees for the society's flagship conference.

**Campus Partner, Perplexity AI - Campus Strategist Program 2025** Sep - Nov 2025

Vellore Institute of Technology

- Drove campus adoption of Comet (Perplexity's browser) at VIT, onboarding 100+ students within three months through targeted outreach and peer-to-peer engagement.

## PROJECTS

**Microcontroller-Based Real-Time ECG Arrhythmia Detection System** Feb 2026 - Present

Individual Research Project · Vellore Institute of Technology

- Researched ECG signal acquisition and conditioning at 360 Hz via AD8232 analog front-end; as well as building an on-MCU DSP pipeline for baseline wander removal, 50/60 Hz powerline noise suppression, and EMG artifact filtering.

- Developing a modified Pan-Tompkins + lightweight ML classifier (SVM / decision tree) targeting 5 arrhythmia classes (NSR, PVC, AF, LBBB, RBBB) with <100 ms end-to-end latency per beat cycle.
- Targeting sensitivity >97% and accuracy >97% per class against the MIT-BIH Arrhythmia Database; transmitting results wirelessly via BLE / Wi-Fi to a mobile dashboard with real-time alerts.

### **FPGA-Based Morse Code Transceiver with UART Serial Bridge**

Jan 2026 - Present

Personal Project · Verilog / SystemVerilog · Intel Quartus Prime · Altera DE2-115

- Implemented a bidirectional Morse code encoder/decoder on FPGA (Altera DE2-115, 50 MHz clock) with real-time LED output and 700 Hz audio tone generation via PWM.
- Constructed a 9600-baud UART serial bridge between FPGA and PC terminal using a 4-state FSM, achieving sub-1 ms end-to-end character transmission latency.

### **Python-Based Network Traffic Analyser & Anomaly Dashboard**

Mar 2026 - Present

Personal Project · Python · Scapy · Matplotlib / Dash

- Built a Python packet-capture tool (Scapy) classifying live traffic across 5+ protocols (TCP, UDP, ICMP, HTTP, DNS) and flagging flow anomalies using 3-sigma statistical thresholds.

### **Project Sentinel - AI Predictive Maintenance System**

Yantra Central Hackathon '25

Team Kaizen · Role: Content Management & Pitching

- Developed an AI-powered predictive maintenance system forecasting equipment failures and estimating Remaining Useful Life (RUL); achieved >90% anomaly detection accuracy across simulated sensor datasets.

### **Active Vibration Control of a Flexible Beam using Feedback & Frequency-Domain Compensation**

Individual Research Project · Digital Control Systems · Vellore Institute of Technology

- Designing and experimentally validating a low-cost digital active vibration control system for a flexible resonant structure, targeting >80% vibration amplitude reduction; employing discrete-time classical control techniques to suppress resonant modes and improve phase margin by 30-40°.

### **Biomimicry Oil Spill Cleanup System**

BIDFEST, VIT - 1st Place

Role: Communications Head

- Engineered a biomimicry-inspired oil-water separation system achieving >85% separation efficiency in prototype trials, winning 1st place at BIDFEST among 10+ competing teams.

## **AWARDS & ACHIEVEMENTS**

- Multiple MUN Awards across national conferences (2025-2026), including Best Delegate, two High Commendations, and several Honourable Mentions.

## **CERTIFICATIONS**

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|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| - Fundamentals of Digital Design for VLSI Chip Design - <b>L&amp;T EduTech</b>        | - Digital Signal Processing - <b>Academy Europe Open University</b>                                           |
| - Semiconductor Characterization - <b>Arizona State University</b>                    | - Analog Electronics, Certificate of Merit - <b>Sanfoundry</b>                                                |
| - Electrodynamics - <b>Korea Advanced Institute of Science and Technology (KAIST)</b> | - Advanced Drone Technology (Air Taxi) Workshop - <b>Bharat Space Education Research Centre x IISc (2025)</b> |
| - AI Foundations for Everyone - <b>IBM</b>                                            |                                                                                                               |